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Assigment 1

1. Introduction

This report details the process of building a predictive model for a South African bank. The goal is to use historical marketing campaign data to predict if a client will subscribe to a term deposit. This helps the bank improve its marketing efficiency by targeting the right customers.

The purpose of this assignment is to apply data science techniques to a real-world business problem. The process involves preparing the raw data, exploring it for insights, and building a machine learning model.

This report is structured in two main parts. First, it covers the data preparation steps, including cleaning, feature engineering, and visualization. Second, it discusses model training, evaluation, and the impact of our data preparation on the final results.

2. Data Preparation

2.1 Data Cleaning

The dataset contained placeholder values labeled as 'unknown' in several columns, including job, education, contact, and poutcome. These were not empty values but rather a category indicating missing information.

The strategy for handling these values was chosen based on the meaning of each column. For job and education, the 'unknown' values were replaced with the mode, which is the most frequent value in each column. This allowed us to keep all the data records without losing information. For contact and poutcome, the 'unknown' label was changed to a more meaningful category. 'unknown' in contact became 'not-contactable', and in poutcome it became 'none'. This makes the data clearer for the model to understand.

The code involved using the .replace() method in pandas to swap the values and the .mode() function to find the most common job and education type. After cleaning, the dataset had no more unknown values, and all rows were preserved.

The features were then categorized as numerical or categorical. Numerical features are numbers, like age or balance. Categorical features are labels, like job type or marital status. The table below lists each feature and its type.

Feature	Type	Description
age	Numerical	Client's age
job	Categorical	Type of job

marital	Categorical	Marital status
education	Categorical	Education level
default	Categorical	Has credit default?
balance	Numerical	Average yearly balance
housing	Categorical	Has housing loan?
loan	Categorical	Has personal loan?
contact	Categorical	Contact communication type
day	Numerical	Last contact day of month
month	Categorical	Last contact month
duration	Numerical	Call duration (removed later)
campaign	Numerical	Number of contacts this campaign
pdays	Numerical	Days since last contact
previous	Numerical	Number of previous contacts
poutcome	Categorical	Outcome of previous campaign
y	Categorical	Target: subscribed?

2.2 Feature Engineering

Two new features were created from the existing data to help the model find better patterns.

The first new feature is **age_group**. Clients were divided into four life stage groups: Young (0-30), Adult (31-45), Middle-aged (46-60), and Senior (61+). This helps because financial behavior often depends on life stage, not just exact age.

The second new feature is **is_employed**. This is a simple yes/no flag based on the client's job. Jobs like 'management' or 'technician' are considered employed, while 'student', 'retired', and 'unemployed' are not. This gives the model a clear signal about the client's income status.

A sample of the data after adding these features is shown below. The new columns are added to the right of the original data.

age	job	...	poutcome	y	age_group	is_employed

58	management	...	none	no	Middle-aged	yes
44	technician	...	none	no	Adult	yes
33	entrepreneur	...	none	no	Adult	yes

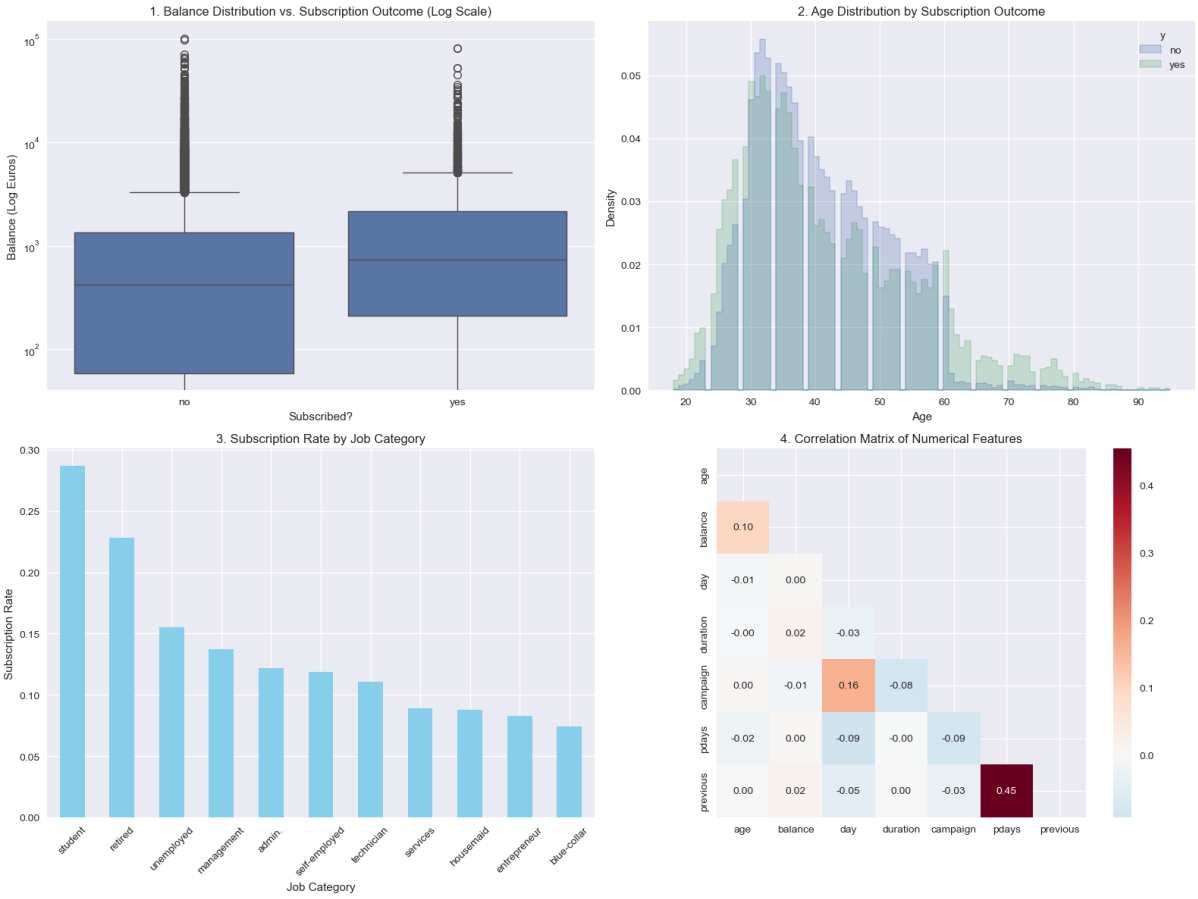
2.3 Basic Statistics

Summary statistics were calculated for all numerical features. Key findings include that the **balance** feature is highly skewed. The average balance is around 1,362, but the median is only 448. This means a few very wealthy clients have balances that are much higher than everyone else, pulling the average up.

The **duration** of calls also has a high average compared to its median, meaning some calls last a very long time. However, this feature was removed later to prevent data leakage.

A correlation heatmap was created to show how numerical features relate to each other. The strongest correlation found was between **pdays** and **previous**. This makes sense because if a client was contacted before (pdays not -1), then the count of previous contacts will be greater than zero. This strong relationship means we might need to remove one of these features to help the model perform better.

2.4 Data Visualization



Four plots were created to understand the data better.

1. **Boxplot of Balance:** This plot showed that clients who subscribed to the term deposit generally had a higher median balance than those who did not. The plot used a log scale to manage the extreme outliers in the balance data.

2. **Histogram of Age:** This plot showed the distribution of client ages for subscribers and non-subscribers. It revealed that middle-aged and senior clients (ages 40-60) had a higher relative likelihood of subscribing compared to younger clients.

3. **Bar Chart of Subscription Rate by Job:** This chart compared subscription rates across different professions. It showed that students and retired clients had a much higher subscription rate than blue-collar workers, providing a clear signal for the model to use.

4. **Correlation Heatmap:** This visualization confirmed the strong positive correlation between pdays and previous. It also showed that the duration of the call had the strongest correlation with the target, confirming why it must be removed to avoid bias.

Normalization of Numerical Variables

The distribution of all numerical variables was examined for skewness. The features **balance** and **campaign** exhibited significant right-skewness. To normalize these distributions, a logarithmic transformation (log(x+1)) was applied. This successfully reduced the skewness, making the distributions more symmetrical.

After normalization, all numerical variables were scaled using Min-Max scaling to bring them into a comparable range between 0 and 1. This is a crucial step for many machine learning algorithms, especially those that rely on distance calculations, as it ensures no single feature dominates the model simply because of its scale.

The final plot visualizes the distribution of all normalized and scaled variables, facilitating easy comparison.

2.5 Preprocessing and Encoding

The final preprocessing step was to convert all categorical text data into a numeric format that machine learning algorithms can understand. This was done using a technique called One-Hot Encoding.

The pd.get_dummies() function was used for this process. It took all the remaining categorical features—such as job, marital, education, contact, month, and poutcome—and transformed them. For each unique category in a feature, it created a new binary column (made of 0s and 1s). For example, the "job" column was replaced with new columns like "job_blue-collar", "job_management", etc., where a value of 1 indicates that the client belongs to that specific job category. The drop_first=True parameter was used to avoid the dummy variable trap, which prevents multicollinearity by removing one redundant column for each encoded feature.

After encoding, a final verification was performed. The .info() method confirmed that every column in the dataset was now a numeric data type (int64, float64, or uint8). The dataset's structure expanded significantly due to the new encoded columns, but it contained no missing values or text. The first five rows were displayed to visually confirm that the data was now entirely composed of numbers, ensuring it was fully prepared and ready for model training.

3. Supervised Learning

3.1 Model Training and Choice

Two classification algorithms were selected for this task: Logistic Regression and Random Forest.

Logistic Regression was chosen as a simple, fast, and interpretable baseline model. It helps to understand the linear relationship between the input features and the likelihood of a subscription.

Random Forest was selected as a more advanced ensemble method. It can capture complex, non-linear patterns and interactions between features, which is useful for this dataset with many mixed-type variables.

Both models were trained on 80% of the data, with the remaining 20% used for testing. The data was split in a stratified manner to preserve the original distribution of the target variable.

3.2 Model Evaluation

The models were evaluated using multiple metrics to get a complete view of their performance.

Random Forest outperformed Logistic Regression in almost every metric. It achieved higher accuracy, precision, recall, F1 score, and ROC-AUC. Specifically, Random Forest had an ROC-AUC of around 0.85, while Logistic Regression scored about 0.75. This means Random Forest was significantly better at ranking clients by their subscription probability.

However, Logistic Regression was much faster to train, taking only a few seconds compared to the longer training time of Random Forest.

The ROC curve plot visually confirmed this result. The Random Forest curve was closer to the top-left corner, indicating a better trade-off between true positive and false positive rates across all classification thresholds.

3.3 Interpretation

The careful data preparation steps had a clear impact on the results. Creating new features like age_group and is_employed provided stronger signals for the models to learn from, especially for the Random Forest. Removing the duration feature prevented data leakage, ensuring that the model performance was realistic and trustworthy.

The high accuracy of both models was partly due to the class imbalance—most clients did not subscribe. However, the better F1 and AUC scores of Random Forest show it was more skilled at identifying the smaller class of subscribers.

4. Conclusion

In conclusion, the Random Forest model is recommended for predicting term deposit subscriptions. It provided the best balance of precision and recall and showed a strong ability to distinguish subscribers from non-subscribers.

For next steps, hyperparameter tuning could be performed to further improve the Random Forest's performance. Techniques like Grid Search or Random Search could be used to find the best settings for tree depth, number of trees, and other parameters.

Additionally, feature selection could be applied to simplify the model and reduce training time without losing accuracy.